SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Correlation and Pattern Analysis

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO4	**Assessment:**	Tool 3– **Correlation and Pattern Analysis**
Weightage:	25%	**Total Marks:**	

CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Student Name:	Sandra Sudevan	**Reg. No.:**	192424422
Section / Batch:	2024	**Date of Submission:**	24/08/26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I Sandra Sudevan(192424422)(Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandra Sudevan

1. **Correlation Analysis of Student Performance**
   Import student datasets containing **attendance, assignment marks, internal marks, study hours, and final marks**. Explore the tables, join relevant datasets, calculate correlations between variables, identify outliers, and use appropriate charts to explain the major patterns in student performance.

Aim

The aim of this analysis is to study how **attendance, assignment marks, internal marks, and study hours** affect students' **final marks**.

1. Dataset Creation

The dataset contains information about 20 students with the following variables:

- **Student ID** – Unique identification number.
- **Attendance** – Percentage of classes attended.
- **Assignment Marks** – Marks obtained in assignments.
- **Internal Marks** – Internal examination marks.
- **Study Hours** – Hours spent studying.
- **Final Marks** – Final examination marks.
-

2. Correlation Analysis

Correlation is used to identify the **relationship between two variables**.

The correlation value ranges from **-1 to +1**:

- **+1** → Strong positive relationship
- **0** → No relationship
- **-1** → Strong negative relationship

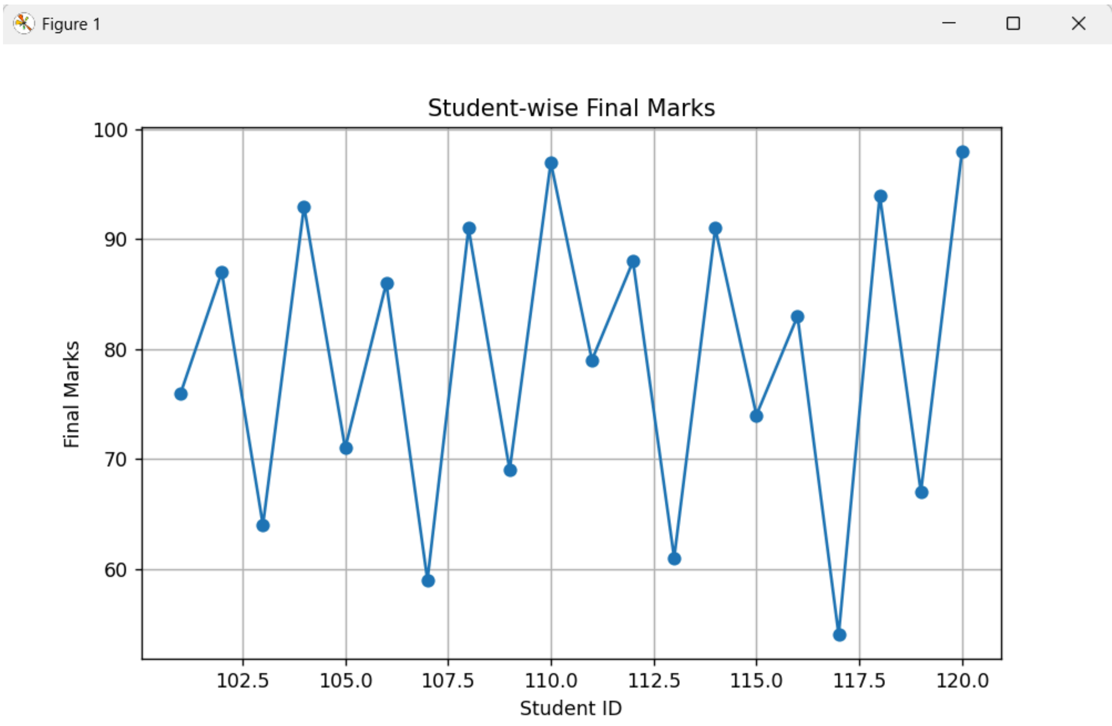
For example, if study hours and final marks have a high positive correlation, it means students who study more hours generally achieve higher final marks.

3. Line Graph

The line graph shows the **final marks of each student**.

It helps us identify:

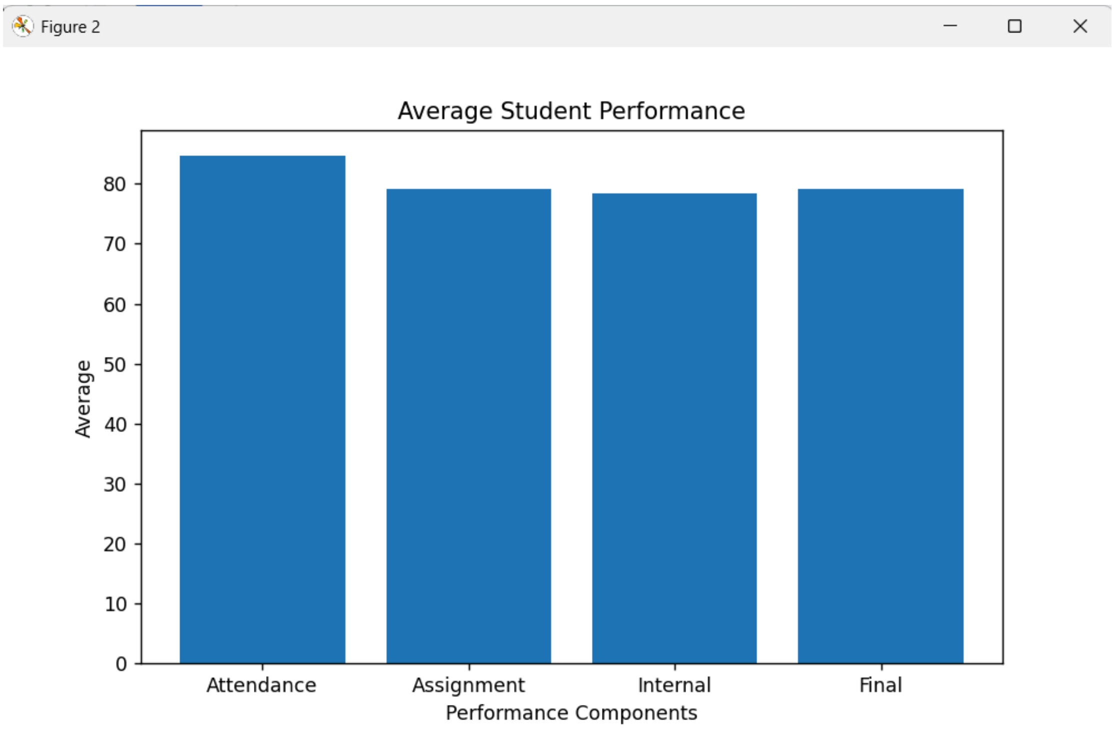
- Students with high marks
- Students with low marks
- Changes in performance between students



4. Bar Graph

The bar graph compares the **average attendance, assignment marks, internal marks, and final marks**.

It gives an overall view of student performance.

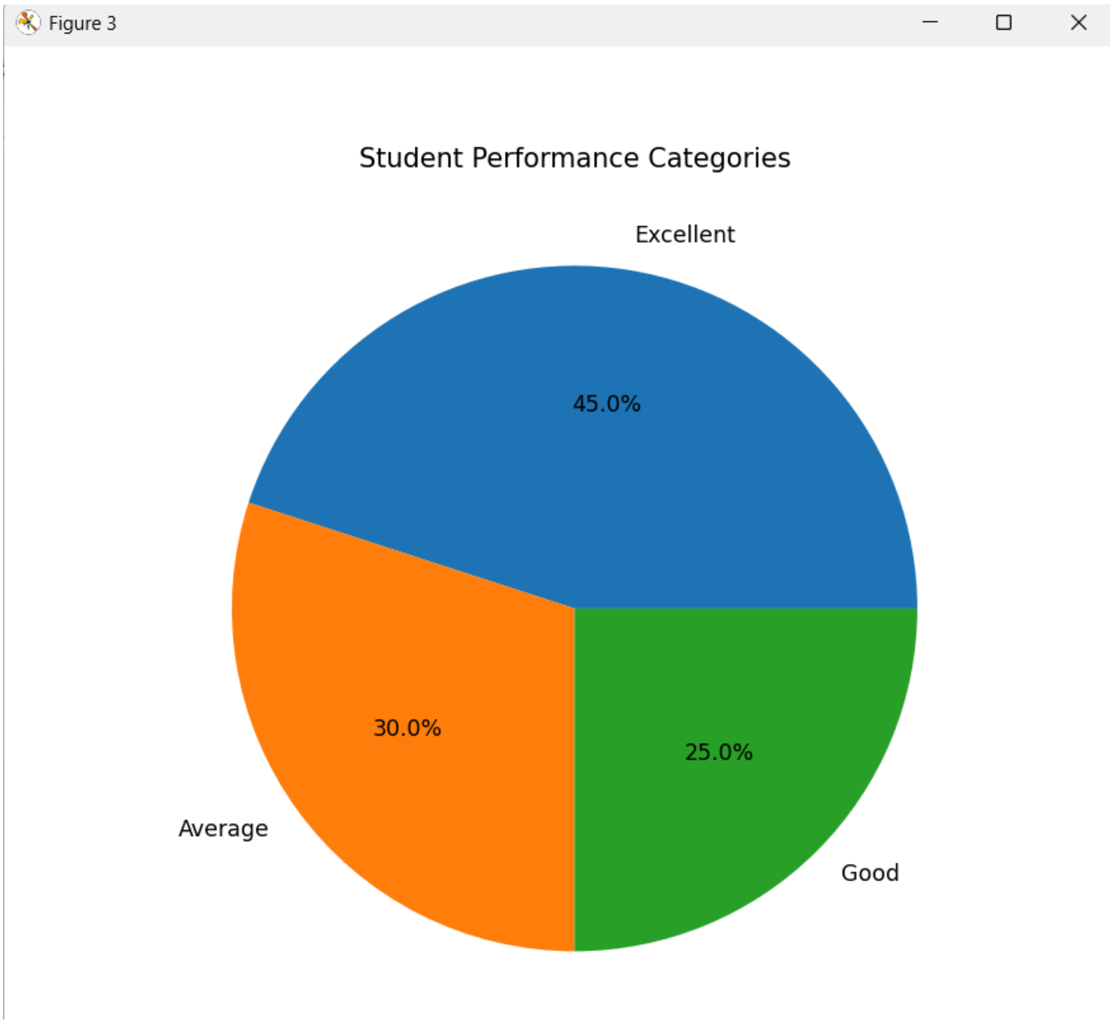


5. Pie Chart

The students are divided into four performance categories:

- **Excellent** – 85 and above
- **Good** – 70–84
- **Average** – 50–69
- **Poor** – Below 50

The pie chart shows the percentage of students in each category.

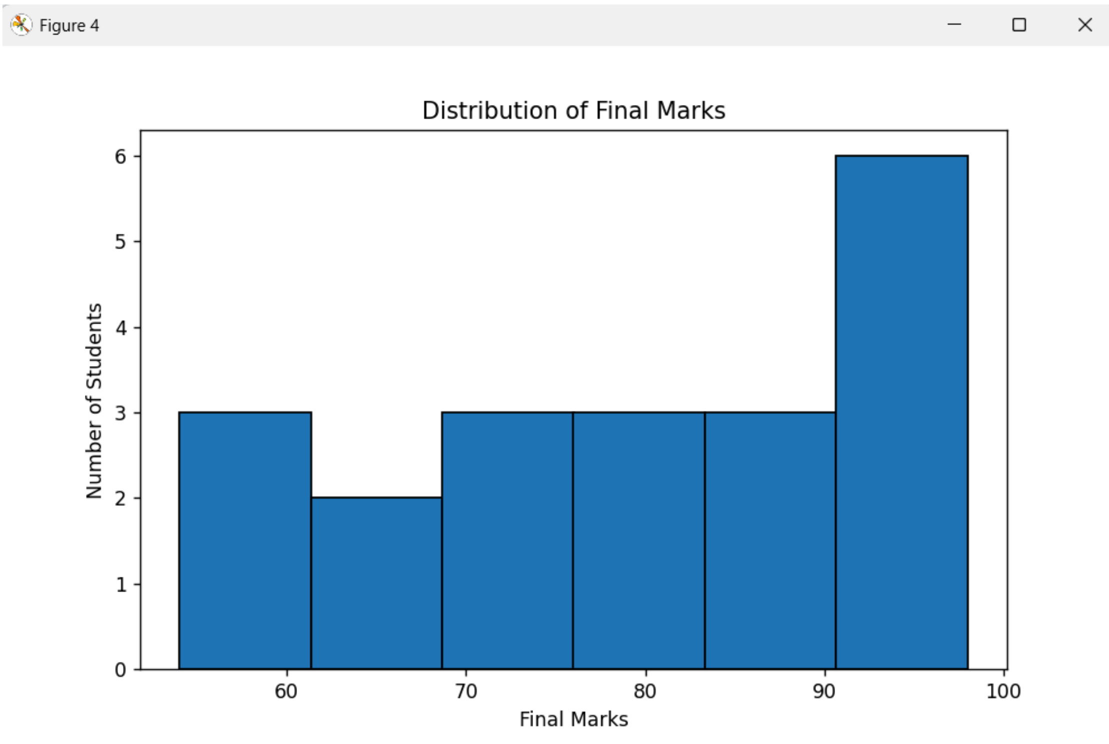


6. Histogram

The histogram shows the **distribution of final marks**.

It helps identify whether most students scored:

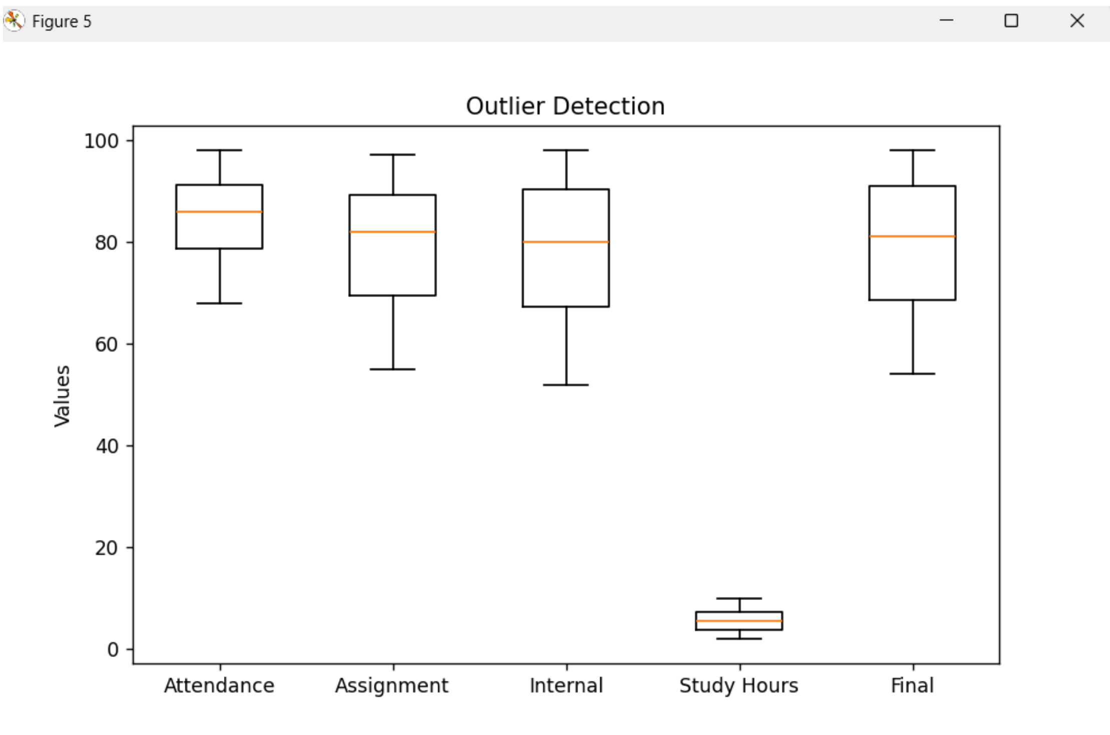
- Low marks
- Medium marks
- High marks



7. Box Plot

The box plot is used to identify **outliers**.

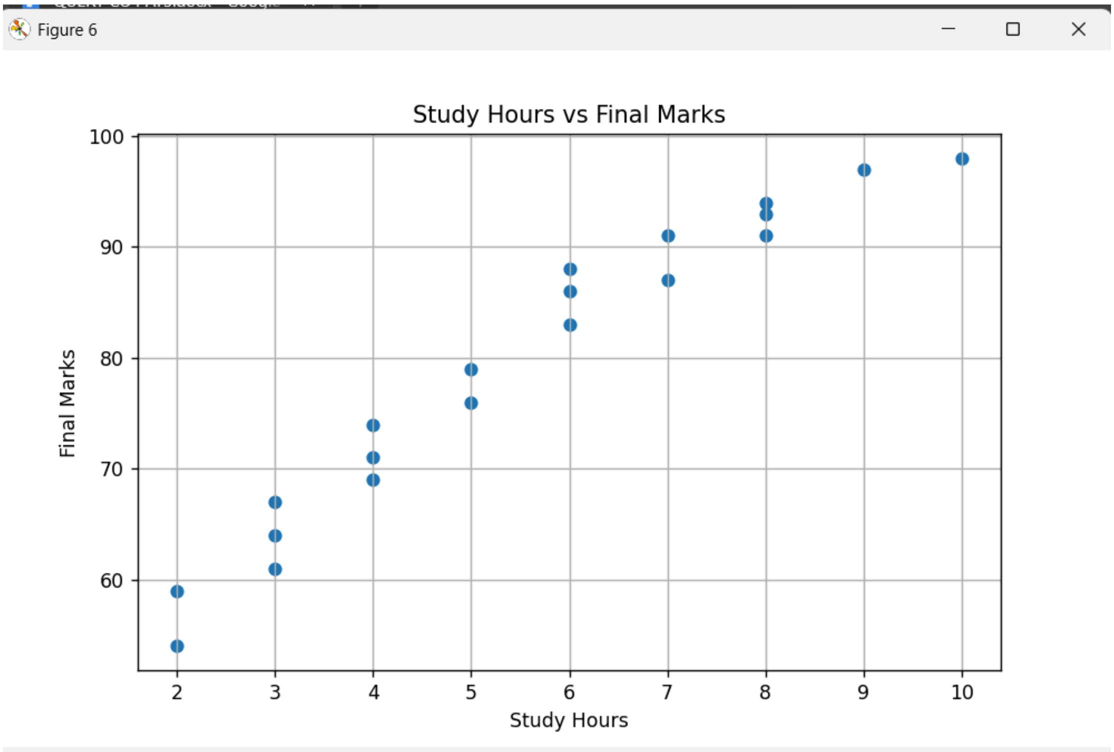
An outlier is a value that is unusually high or low compared with the other observations.



8. Study Hours vs Final Marks

The scatter graph compares **study hours with final marks**.

If the points generally move upward, it indicates a **positive relationship** between study time and academic performance.



Conclusion

The analysis helps identify the major factors affecting student performance. **Attendance, assignment marks, internal marks, and study hours can be compared with final marks using correlation and graphs.** Students with better academic participation and more study time generally tend to show better final performance. The visualizations also make it easier to identify performance groups and unusual values.

2. **Sales and Customer Pattern Analysis**
   Import sales and customer datasets and combine them using suitable table-joining techniques. Analyze the correlation between **customer age, purchase frequency, discount, and purchase amount**. Create meaningful customer groups, identify unusual purchasing patterns, and present the results using suitable visualizations.

Aim

The aim of this analysis is to study **customer purchasing behavior** by analyzing age, purchase frequency, discount, and purchase amount.

1. Dataset Creation

Two datasets are used:

Customer Dataset

- Customer ID
- Age

Sales Dataset

- Customer ID
- Purchase Frequency
- Discount
- Purchase Amount

The two datasets are combined using **Customer ID**.

2. Table Joining

The customer and sales tables are joined using an **inner join**.

This creates one combined dataset containing both customer information and their purchasing details.

3. Correlation Analysis

Correlation is calculated between:

- **Age and Purchase Amount**
- **Age and Purchase Frequency**
- **Discount and Purchase Amount**
- **Purchase Frequency and Purchase Amount**

The correlation value ranges from **-1 to +1**.

- Positive value → variables increase together.
- Negative value → one variable increases while the other decreases.
- Value near 0 → weak or no relationship.

4. Age vs Purchase Amount

The graph shows whether **customer age affects spending**.

It helps identify whether younger, middle-aged, or older customers tend to spend more.

5. Purchase Frequency vs Purchase Amount

This graph compares how often customers purchase with how much they spend.

A positive relationship indicates that customers who purchase more frequently may also have higher total purchase amounts.

6. Discount vs Purchase Amount

This graph studies whether discounts influence customer spending.

It helps identify whether customers receiving higher discounts tend to make larger purchases.

7. Customer Grouping

Customers are divided into three groups based on age:

- **Young** → Below 30
- **Middle-aged** → 30–49
- **Senior** → 50 and above

This makes it easier to compare purchasing behavior between different age groups.

8. Outlier Detection

A **box plot** is used to identify unusual purchase amounts.

An outlier may represent a customer who spent an unusually large or small amount compared with other customers.

9. Unusual Purchasing Patterns

The IQR method is used to identify unusual purchase values.

The formula is:

IQR = Q3 − Q1

Values below:

Q1 − 1.5 × IQR

or above:

Q3 + 1.5 × IQR

are considered potential outliers.

Conclusion

The analysis helps understand **customer purchasing patterns and spending behavior**. By combining customer and sales data, correlation analysis, customer grouping, and outlier detection can reveal which factors are associated with higher purchase amounts and identify unusual purchasing behavior.

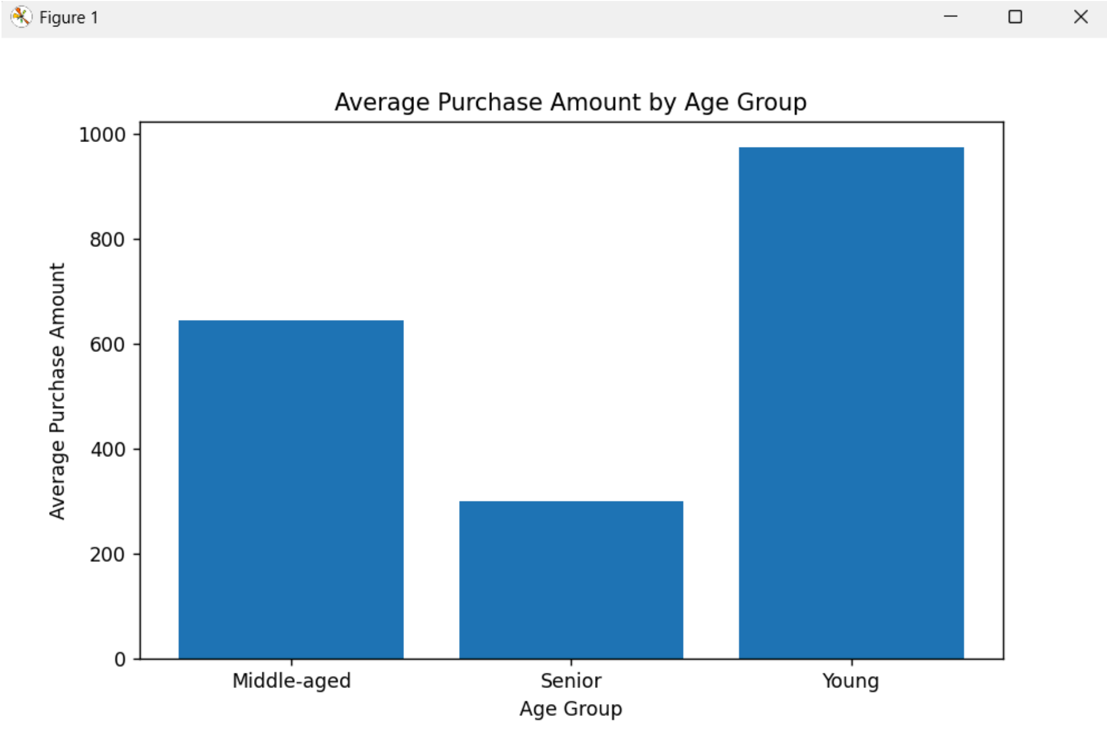


Figure 1
Average Purchase Amount by Age Group
Average Purchase Amount
0
200
400
600
800
1000
Middle-aged
Senior
Young
Age Group

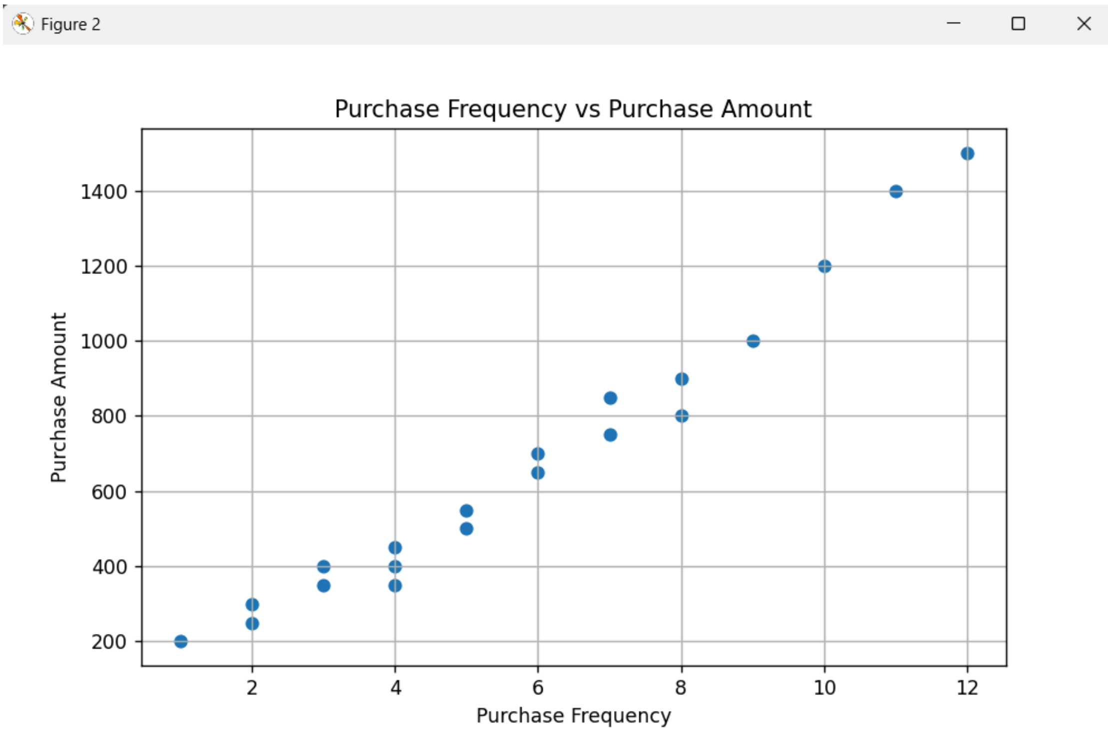


Figure 2
Purchase Frequency vs Purchase Amount
Purchase Amount
200
400
600
800
1000
1200
1400
2
4
6
8
10
12
Purchase Frequency

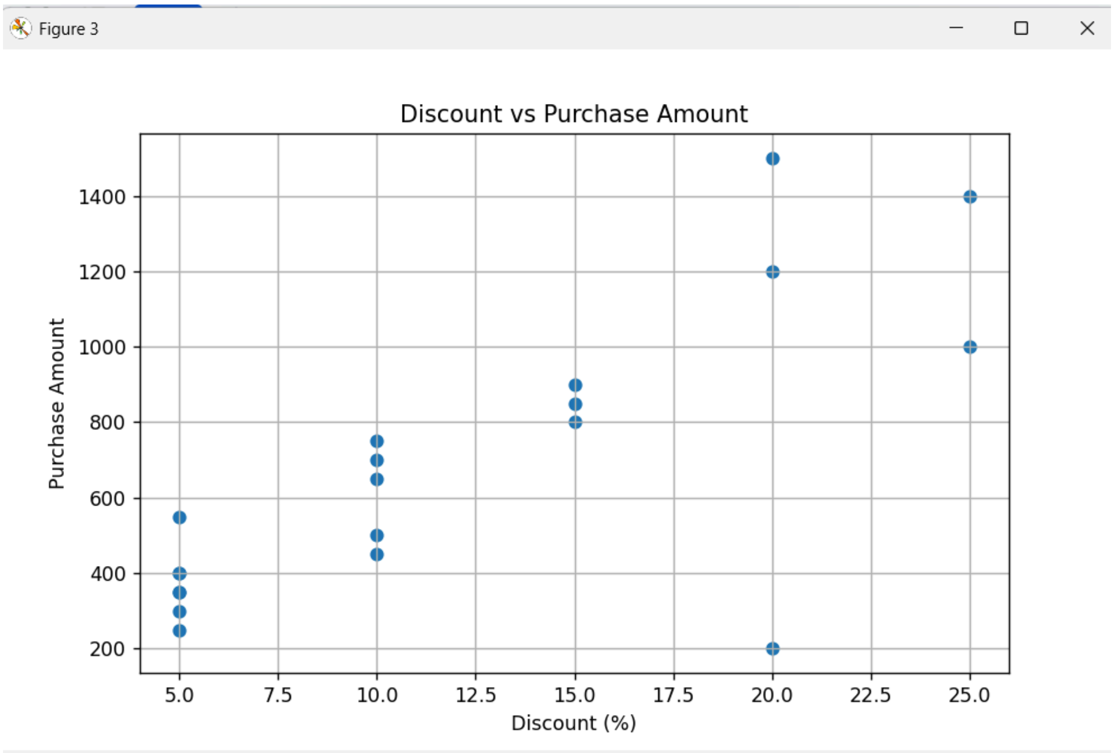


Figure 3
Discount vs Purchase Amount
Purchase Amount
Discount (%)
1400
1200
1000
800
600
400
200
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0

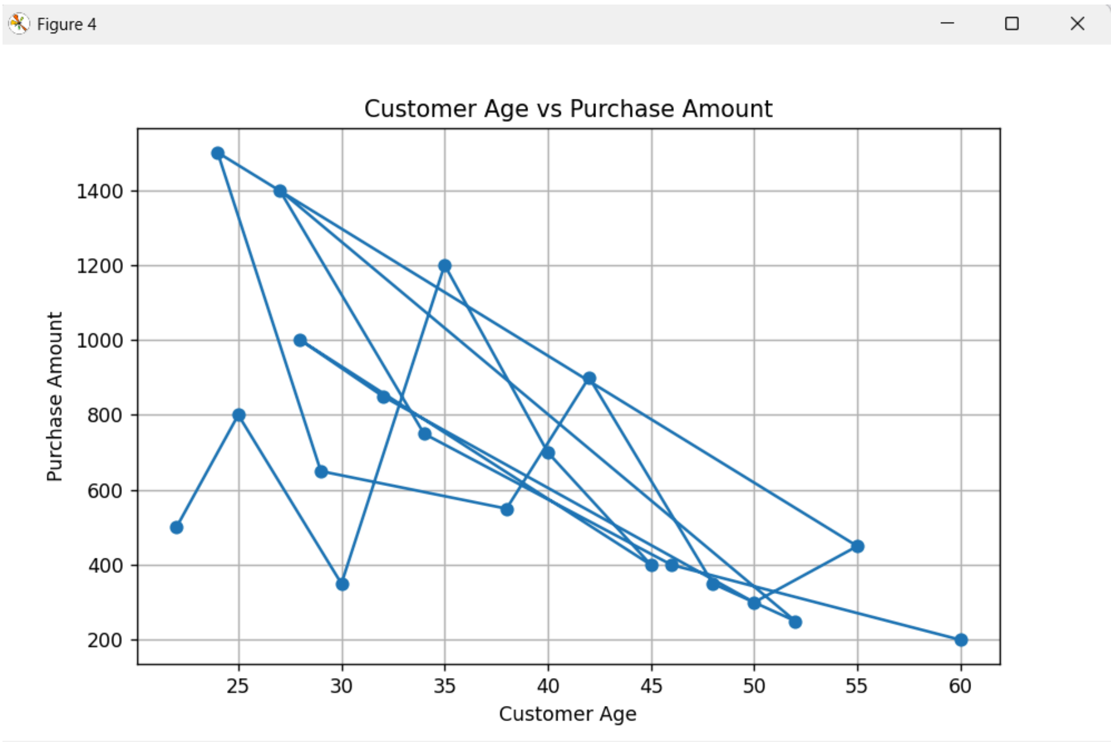


Figure 4
Customer Age vs Purchase Amount
Purchase Amount
Customer Age
1400
1200
1000
800
600
400
200
25
30
35
40
45
50
55
60

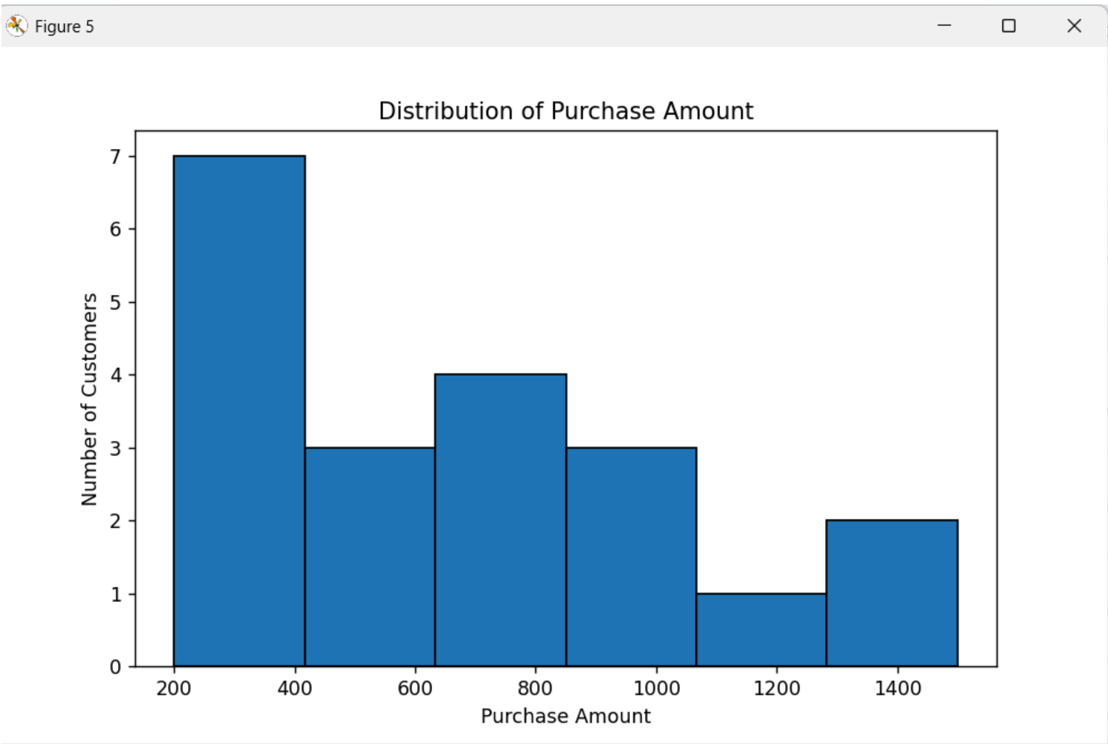


Figure 5
Distribution of Purchase Amount
Number of Customers
Purchase Amount
0
1
2
3
4
5
6
7
200
400
600
800
1000
1200
1400


Figure 6
Purchase Amount Outlier Detection
Purchase Amount
200
400
600
800
1000
1200
1400
1